Type MS · Volume of revolution about the y-axis

Answer 1 of 4 · Question 4 part b · 4 marks

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4(a)	**Mark parts (i) and (ii) together.** $2y-x-4=0 \Rightarrow y=\frac{x+4}{2}$ oe $\frac{x^2}{4}+2=\frac{x+4}{2}$ oe $\Rightarrow \frac{x^2+8}{4}=\frac{x+4}{2}$ oe $\Rightarrow x^2+8=2x+8$ oe $\Rightarrow x^2-2x=0 \Rightarrow x(x-2)=0$ oe $\Rightarrow x=(0,)\ 2$ $\Rightarrow y=2,\ 3$ So coordinates at point A are (0, 2) * and at point B (2, 3)	M1 M1 A1* cso A1 [4]
	NB SC1 – for correct and complete substitution of $x = 0$ into the equation of curve S or line l to show that $y = 2$, where there is no other work, other work is incorrect or no marks would otherwise be gained. Award this as 1st M mark. THERE ARE NO OTHER MARKS AVAILABLE FOR PART (a) IF THIS IS ALL STUDENTS DO	
(b)	$V=\pi\int_{2}^{"3"}(4y-8)\,dy-\pi\int_{2}^{"3"}(2y-4)^2\,dy$ $\left[\Rightarrow \pi\int_{2}^{"3"}(-4y^2+20y-24)\,dy \quad \text{or} \quad \pi\int_{2}^{"3"}(4y-8)\,dy-\pi\int_{2}^{"3"}(4y^2-16y+16)\,dy\right]$ $=(\pi)\left[-\frac{4y^3}{3}+\frac{20y^2}{2}-24y\right]_{("2")}^{("3")}$ or $(\pi)\left[\frac{4y^2}{2}-8y\right]_{("2")}^{("3")}-(\pi)\left[\frac{4y^3}{3}-\frac{16y^2}{2}+16y\right]_{("2")}^{("3")}$ $=(\pi)\left[\left(\frac{4\times"3"^3}{3}+\frac{20\times"3"^2}{2}-24\times"3"\right)-\left(\frac{4\times"2"^3}{3}+\frac{20\times"2"^2}{2}-24\times"2"\right)\right]$ $=\frac{2}{3}\pi$ **ALT** $V=\pi\int_{2}^{"3"}(4y-8)\,dy-\pi\times\frac{1}{3}\times2^2\times1$ $=\pi\left[\frac{4y^2}{2}-8y\right]_{"2"}^{"3"}\left(-\frac{4}{3}\pi\right)$ $=\pi\left[\left(2\times"3"^2-8\times"3"\right)-\left(2\times"2"^2-8\times"2"\right)\right]\left(-\frac{4}{3}\pi\right)$ $=\frac{2}{3}\pi$	M1 M1 M1 A1 [4] M1 M1 M1 A1
		Total 8 marks

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Part	Mark	Notes
(a)	M1	Correctly equates the equation of S with the equation of l
	M1	Forms a quadratic and a minimally acceptable attempt to solve their quadratic. For a 2TQ the **solution** of their quadratic for must be correct, **but any zero solution doesn't need to be shown as it's obvious from the given diagram.** If they achieve a 3TQ and attempt to solve – see general guidance.
	A1* cso	For correct substitution of $x = 0$ to show coordinates of A (0, 2) or $x = 0$, $y = 2$. Allow coordinates without brackets. SC1 (see MS) if this is the only work done.
	A1	For the coordinates of B (2, 3) $x = 2$, $y = 3$ Allow coordinates without brackets.
Note on labelling: **If the labels (i) and (ii) are not present, the marks can be awarded if** • **The coordinates appear in the correct order or** • **They are labelled with A and B** **If there is ambiguity award A1 A0 if both correct and A0 A0 if only one is correct**		
(b)	M1	For a correct statement for the volume of rotation with 2 and their upper limit used correctly and correctly including π. The lower limit **must** be 2.
	M1	For a minimally acceptable attempt to integrate (see general guidance) **and** no power of y must decrease. π and limits do not need to be present to gain this mark. There must be a minimum of 2 terms to integrate.
	M1	For substituting their limits into their changed expression the correct way round. There must be at least one clear substitution of each limit. π does not need to be present to gain this mark.
	A1	For the correct final volume $= \frac{2}{3}\pi$ accept $0.\dot{6}\pi, 0.67\pi$ or better
	ALT	
	M1	For a correct statement for the volume of rotation of the curve with 2 and their upper limit used correctly including π minus the correct formula used for the volume of a cone. The lower limit **must** be 2. **This mark may also be awarded if the integral is subtracted the wrong way round.**
	M1	For a minimally acceptable attempt to integrate (see general guidance) **and** no power of y must decrease. π, limits and $-\frac{4}{3}\pi$ do not need to be present to gain this mark. There must be a minimum of 2 terms to integrate.
	M1	For substituting their limits into their changed expression. There must be at least one clear explicit substitution of each limit. This mark can be implied by a correct final answer. π and $-\frac{4}{3}\pi$ do not need to be present to gain this mark.
	A1	For the correct final volume $= \frac{2}{3}\pi$ accept $0.\dot{6}\pi, 0.67\pi$ or better **If a negative value is found and changed at the end to a positive value, this final A mark cannot be awarded.**
	Note: any candidate incorrectly rotating around x axis, 2nd & 3rd M marks are available for integrating an expression & substituting in. Max mark M0 M1 M1 A0.	

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Question	Scheme	Marks

Type MS · Volume of revolution about the y-axis

Answer 2 of 4 · Question 9 part whole · 9 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

9	Intersections of curve and line: $x^2-2=-2x+13$ $\quad y=\left(\frac{y-13}{2}\right)^2-2$ $\Rightarrow x^2+2x-15=0$ $\Rightarrow (x+5)(x-3)=0$ or $\Rightarrow y^2-30y+161=0$ $\Rightarrow x=(-5),\ 3$ $\quad \Rightarrow (y-7)(y-23)=0$ when $x=3, y=7$ $\quad \Rightarrow y=7,\ (23)$ <u>Volume of cone</u> Using formula: Volume of cone $=\frac{1}{3}\times\pi\times3^2\times(13-7)=18\pi$ or using integration: Volume of cone $=\pi\int_7^{13}\left(\frac{13}{2}-\frac{y}{2}\right)^2\,dy=\frac{\pi}{4}\left[169y-13y^2+\frac{y^3}{3}\right]_7^{13}=18\pi$ <u>Volume of solid generated by rotating the curve:</u> When $x=0, y=-2$ Volume $=\pi\int_{-2}^{7}(y+2)\,dy=\pi\left[\frac{y^2}{2}+2y\right]_{-2}^{7}$ $=\pi\left\{\left(\frac{49}{2}+14\right)-\left(\frac{(-2)^2}{2}+(-4)\right)\right\}=\frac{81}{2}\pi$ Total volume $=18\pi+\frac{81}{2}\pi=\frac{117}{2}\pi$ oe	M1 M1 A1 B1 M1M1 ddM1 A1 A1 [9]
		Total 9 marks

Type MS · Volume of revolution about the y-axis

Answer 2 of 4 · Question 9 part whole · 9 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

	M1	Attempt to solve $y+2x=13$ with $y=x^2-2$ and obtain a three term quadratic in x or y. ($x^2+2x-15=0$ or $y^2-30y+161=0$) (Condone = 0 missing.)
	M1	Attempt to solve (see general principles for marking quadratics).
	A1	$y=7$
	B1	18π correct volume of cone, in terms of π.
	M1	Correct integral for volume generated by rotating curve. Volume $=\pm\pi\int_{(-2)}^{(7)}(y+2)\ (\mathrm{d}y)$ Ignore limits and condone dy omitted or with the wrong variable.
	M1	Evidence of completing integration (see general principles for integration). Ignore limits. π may be missing. If $y+2$ is combined with another expression in an attempt to add or subtract two volumes, you may accept evidence of integration for the combined expression.
	dd M1	Dep on previous two M marks. Substitute both correct limits $y=-2$ and $y=7$ into the result of their integration. The full substitution should be shown unless the answer is $\frac{81}{2}\pi$ oe.
	A1	$\frac{81}{2}\pi$ oe, given in terms of π. A correct answer from the correct integration implies M1 for a correct substitution of limits.
	A1	$\frac{117}{2}\pi$ oe, correct total volume, in terms of π.

Type MS · Volume of revolution about the y-axis

Answer 2 of 4 · Question 9 part whole · 9 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Volume of revolution about the y-axis

Answer 3 of 4 · Question 7 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part whole in this mark scheme extract

7	$x=0,\ y=0^2+2=2$ $\left[y=x^2+2\Rightarrow x^2=y-2\right]$	B1
	$V=\pi\int_2^a x^2\ \mathrm{d}y\Rightarrow 18\pi=\pi\int_2^a (y-2)\ \mathrm{d}y,=\pi\left[\frac{y^2}{2}-2y\right]_2^a$	M1,M1
	$18\pi=\pi\left[\left(\frac{a^2}{2}-2a\right)-\left(\frac{2^2}{2}-2\times 2\right)\right]$	M1
	$0=a^2-4a-32$	M1A1
	$a^2-4a-32=(a+4)(a-8)=0\Rightarrow a=8$	M1A1 [8]
	ALT $x=0,\ y=0^2+2=2$ $\left[y=x^2+2\Rightarrow x^2=y-2\right]$	[B1
	$18\pi=\pi\int_2^a (y-2)\ \mathrm{d}y,\ \ 18\pi=\pi\left[\frac{(y-2)^2}{2}\right]_2^a$	M1,M1
	$18\pi=\pi\left[\left(\frac{a-2}{2}\right)^2-\left(\frac{2-2}{2}\right)^2\right]$	M1
	$18=\left(\frac{a-2}{2}\right)^2\Rightarrow 36=(a-2)^2$	M1A1
	$a=2\pm\sqrt{36}\Rightarrow a=8$	M1A1]
		Total 8 marks

Type MS · Volume of revolution about the y-axis

Answer 3 of 4 · Question 7 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part whole in this mark scheme extract

Question	Notes	Marks
7	The intersection of S with the y-axis is at the point with coordinates (0, 2). May be given as when $x = 0, y = 2$	B1
	For a correct statement for the volume, condone missing or incorrect limits and missing π $\left[y = x^2 + 2 \Rightarrow x^2 = y - 2\right]$ $V = \pi\int_2^a x^2 \, dy \Rightarrow 18\pi = \pi\int_2^a (y-2) \, dy$	M1
	METHOD A For an attempt to integrate the expression which must be in the minimally acceptable form $y \pm 2$ $18\pi = \pi\left[\frac{y^2}{2} - 2y\right]_2^a$ [Ignore limits and π for this mark]	M1
	For substituting the limits into their integrated expression to form an equation in a. No simplification is required for this mark. $18\pi = \pi\left[\left(\frac{a^2}{2} - 2a\right) - \left(\frac{2^2}{2} - 2\times 2\right)\right]$	M1
	For forming a 3TQ in terms of a $\quad 0 = a^2 - 4a - 32$	M1
	For the correct 3TQ $0 = a^2 - 4a - 32$	A1
	For an attempt to solve their 3TQ $a^2 - 4a - 32 = (a+4)(a-8) = 0 \Rightarrow a = ...$	M1
	For the correct value of $a = 8$ If $a = -4$ is also stated then it must be rejected.	A1 [8]
	METHOD B For an attempt to integrate the expression which must be in the minimally acceptable form $y \pm 2$ $18\pi = \pi\left[\frac{(y-2)^2}{2}\right]_2^a$ [Ignore limits and π for this mark]	M1
	For substituting the limits into their integrated expression to form an equation in a. No simplification is required for this mark. $18\pi = \pi\left[\left(\frac{a-2}{2}\right)^2 - \left(\frac{2-2}{2}\right)^2\right]$	M1
	For forming an equation in a $\quad 18 = \left(\frac{a-2}{2}\right)^2 \Rightarrow 36 = (a-2)^2$	M1A1
	For an attempt to solve their equation. $a = 2 \pm \sqrt{36} \Rightarrow a = 8$	M1A1

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Answer 3 of 4 · Question 7 part whole · 8 marks

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	Total 8 marks

Type MS · Volume of revolution about the y-axis

Answer 3 of 4 · Question 7 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Volume of revolution about the y-axis

Answer 4 of 4 · Question 6 part b · 7 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

6(a)	$2x^2 = \frac{1}{4x} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}$ $y = 2\times\left(\frac{1}{2}\right)^2 = \frac{1}{2} \Rightarrow \left(\frac{1}{2}, \frac{1}{2}\right)$	M1A1 B1 [3]
(b)	$y = 2x^2 \Rightarrow x^2 = \frac{y}{2}$, $y = \frac{1}{4x} \Rightarrow x^2 = \frac{y^{-2}}{16}$ or $\left(\frac{1}{4y}\right)^2$ o.e. $V = \pi\int_{0.5}^{4}\left(\frac{y}{2}\right)\mathrm{d}y - \pi\int_{0.5}^{4}\left(\frac{y^{-2}}{16}\right)\mathrm{d}y$ $V = \pi\left[\frac{y^2}{4}\right]_{0.5}^{4} - \pi\left[\frac{y^{-1}}{-16}\right]_{0.5}^{4}$ $V = \frac{\pi}{4}\left(4^2 - \left(\frac{1}{2}\right)^2\right) + \frac{\pi}{16}\left(\frac{1}{4} - \frac{1}{0.5}\right) = \frac{245\pi}{64}$ [The decimal equivalent of the area is 12.0264….]	B1,B1 M1 M1A1 M1A1 [7]
		Total 10 marks

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Answer 4 of 4 · Question 6 part b · 7 marks

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		Ignore limits and π
	M1	For substituting their limits the correct way around into their integrated expression. You must see this if their integrated expression is incorrect, or their limits are incorrect. If the integration and limits are all correct, the correct volume seen (either in exact or in decimal form i.e. 12.0….) scores this mark. Accept partly processed, for example $\frac{\pi}{4}\left(16-\frac{1}{4}\right)+\frac{\pi}{16}\left(\frac{1}{4}-\frac{1}{0.5}\right)$ as long as you can see **four** calculations/terms as above. Ignore π for this mark.
	A1	For $\frac{245\pi}{64}$
	SC rotates around the x axis. Maximum score is B0B0M1M1A0M1A0	
	B0B0	Not available
	M1	For a correct expression for the volume with the correct calculated limits for x $\left(\sqrt{2}\right.$ and $\left.\frac{1}{16}\right)$ and π Accept the **[correct only]** expressions either way around. You may see π added in at the end. That is fine and please award this mark if that is the case. $V=\pi\int_{\frac{1}{16}}^{\sqrt{2}}\left(\frac{x^{-1}}{4}\right)^2\,\mathrm{d}x-\pi\int_{\frac{1}{16}}^{\sqrt{2}}\left(2x^2\right)^2\,\mathrm{d}x$
	M1	For an attempt to integrate one of their two only expressions. $V=\pi\left[-\frac{x^{-1}}{16}-\frac{4x^5}{5}\right]_{\frac{1}{16}}^{\sqrt{2}}$ Ignore π and limits for this mark
	A0	Not available
	M1	For substituting their values into the integrated expression correctly the correct way around. $V=\pi\left[\left(-\frac{\left(\sqrt{2}\right)^{-1}}{16}-\frac{4\left(\sqrt{2}\right)^5}{5}\right)-\left(-\frac{\left(\frac{1}{16}\right)^{-1}}{16}-\frac{4\left(\frac{1}{16}\right)^5}{5}\right)\right]$
	A0	Not available

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Answer 4 of 4 · Question 6 part b · 7 marks

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Question	Scheme	Marks